



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division
MTech in Artificial Intelligence and Machine Learning
S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS13 – Communication Networks – [Weightage 12%]

Read through this entire document very carefully before you start!

Problem Statement

In a computer communication network, routers are interconnected through links with varying transmission latency costs. During high network traffic conditions, the routing agent is required to identify a minimum-latency path between a given source router and destination router while minimizing the overall transmission latency. Each node in the graph represents a router and each edge represents a communication link associated with a latency value. Artificial ants traverse the network and collaboratively discover efficient routes by depositing pheromones on promising paths. Over multiple iterations, the agent should learn and reinforce paths with lower communication costs. For heuristic design, consider the inverse of edge latency as the heuristic desirability for path selection. Dynamically accept the source and destination nodes during execution.

Use the following Ant Colony Optimization configurations to identify the minimum-latency path and analyze the behavior of the routing agent.

- Explain the PEAS (Performance Measure, Environment, Actuators, Sensors) for the routing agent.
- Implement the Ant Colony Optimization algorithm using Python. Use the below combinations of hyperparameters and interpret your observations.

Parameters

- Initial pheromone value = 1.0
- Number of iterations = 100
- Q (pheromone constant) = 100

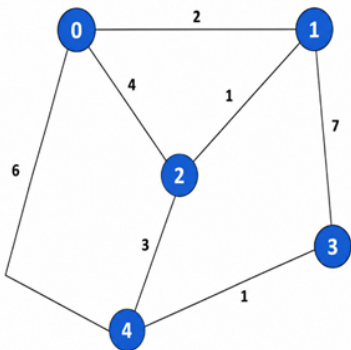
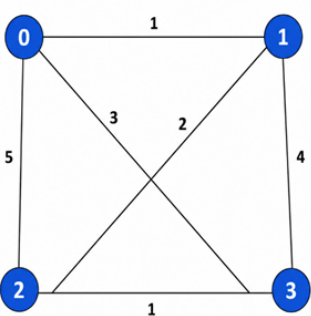
Scenario 1

- Number of Ants = 10
- Relative importance placed on pheromones (α) = 1.0
- Relative importance placed on heuristic information (β) = 2.0
- Evaporation rate (ρ) = 0.5

Scenario 2

- Number of Ants = 10
 - Relative importance placed on pheromones (α) = 2.5
 - Relative importance placed on heuristic information (β) = 1.0
 - Evaporation rate (ρ) = 0.3
- c. Compare the performance of your ACO algorithm under both scenarios and with a deterministic shortest path algorithm (e.g., Dijkstra's Algorithm).
Your analysis should include:
- Difference in convergence speed between the two scenarios
 - Comparison of ACO results with the optimal shortest path
 - Effect of parameter settings (α , β , ρ) on solution quality

Sample Input

Case 1 Nodes (Routers): 0, 1, 2, 3, 4 Edges (Links): 0 – 1: 2 0 – 2: 4 1 – 2: 1 1 – 3: 7 2 – 4: 3 3 – 4: 1 0 – 4: 6	Case 2 Nodes (Routers): 0, 1, 2, 3 Edges (Links): 0 – 1: 1 1 – 2: 2 0 – 2: 5 2 – 3: 1 1 – 3: 4 0 – 3: 3
	

Sample Output

Case 1 Best Path: 0 → 1 → 2 → 4 Minimum Latency: 6, Iterations: 45	Case 2 Best Path: 0 → 1 → 2 → 3 Minimum Latency: 4, Iterations: 52
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Note: Multiple optimal paths may exist with the same minimum latency.

Display the output in outputPS13.txt.

Deliverables:

- PDF document **designPSXX <group id>.pdf** detailing your solution.
- **[Group id]_Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- A single .py file containing the complete Python implementation. Do not fragment your code into multiple files/modules.
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id]_A1_PSXX_XXXXXXXXXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.
 - XXXXXXXXXXXX refers to the title/topic of the problem statement. For PS13, the title of the problem statement is “Communication Networks”. Therefore, the zip file name can be: **G026_A1_PS13_CommunicationNetworks.zip**

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure proper error handling for invalid graph inputs, disconnected nodes, and malformed input files.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries in [this dedicated discussion section](#). Beware that only hints will be provided and queries asked in **other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **8th June 2026 11:55 PM IST (Monday)**.
2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 8th June 2026 11:56PM to 9th June 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 9th June 2026 11:56PM to 10th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6 marks.
 - c. Beyond 10th June 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id]_A1_PSXX_XXXXXXXXXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means such as email, etc., will not be graded.

Evaluation

- The assignment carries **12 Marks**.
- Grading will depend on:
 - Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.